

pyDMS

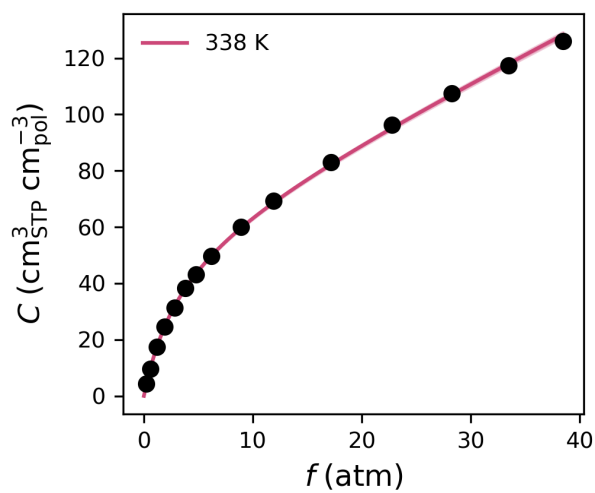
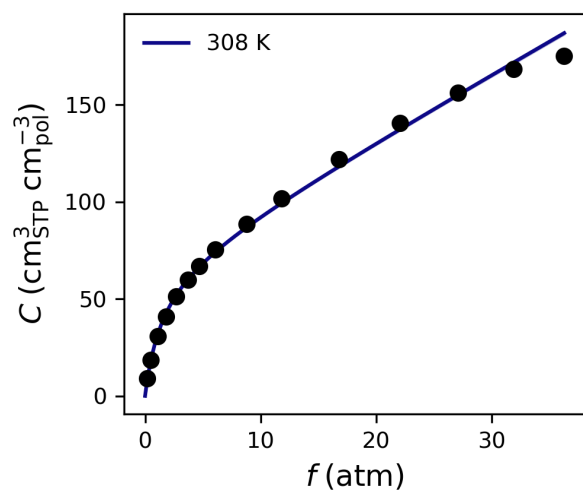
Reproducible Dual-Mode Sorption Parameters

Analysis performed at 2025-06-13 15:44:38 Eastern Daylight Time

DMS Parameters

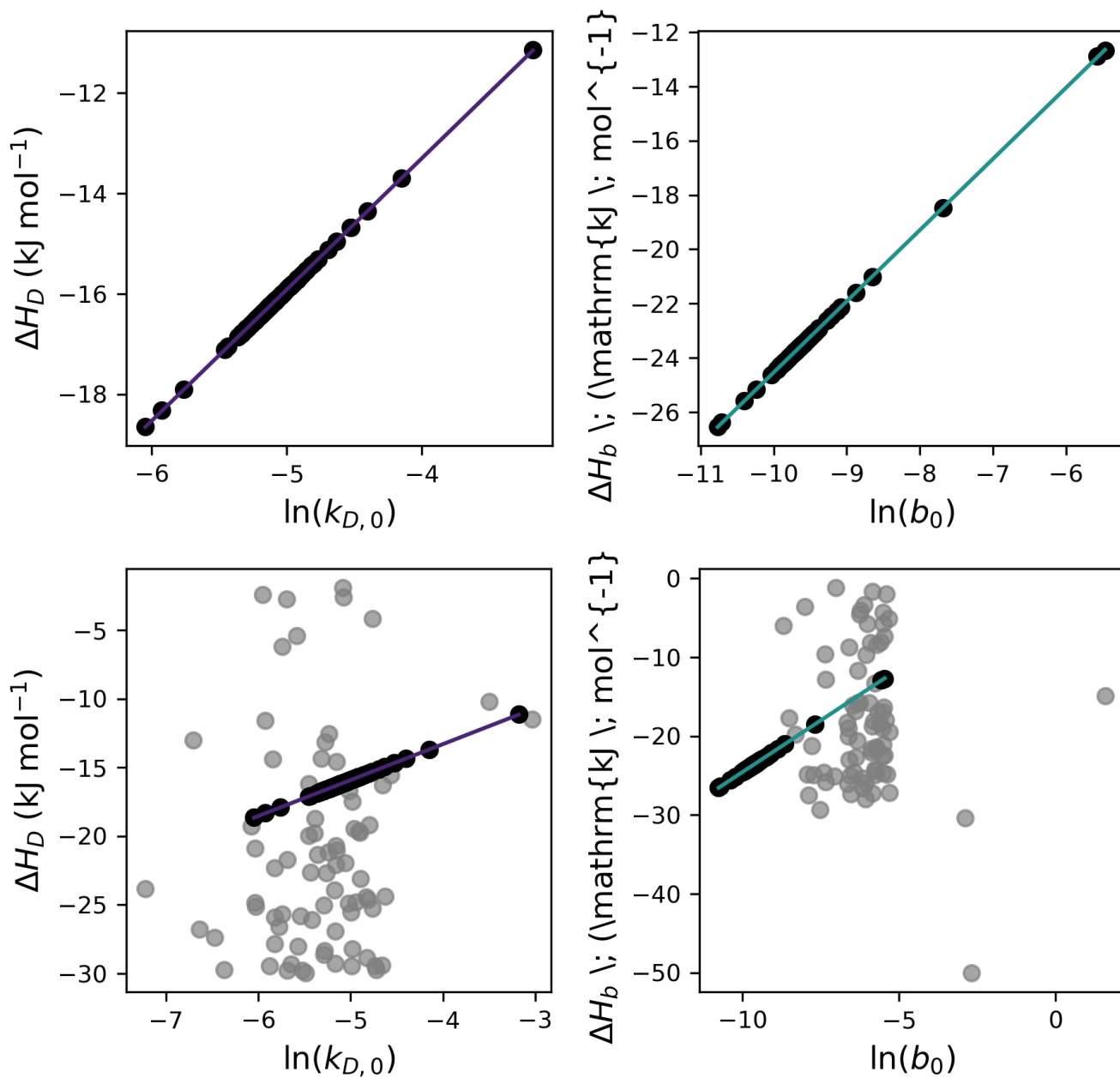
Temperature	Parameter	Result $\pm$ Error
308 K	k_D	$3.3786318649081175 \pm 0.02691992881499278$
	C'_H	$67.11576344281725 \pm 0.49492409907401225$
	b	$0.6530777217639311 \pm 0.004087053248861333$
338 K	k_D	$1.9267688843868973 \pm 0.04862234349979731$
	C'_H	$58.96776275520463 \pm 0.6955545772754874$
	b	$0.2881647198833921 \pm 0.0026434412199819062$

Sorption Isotherms

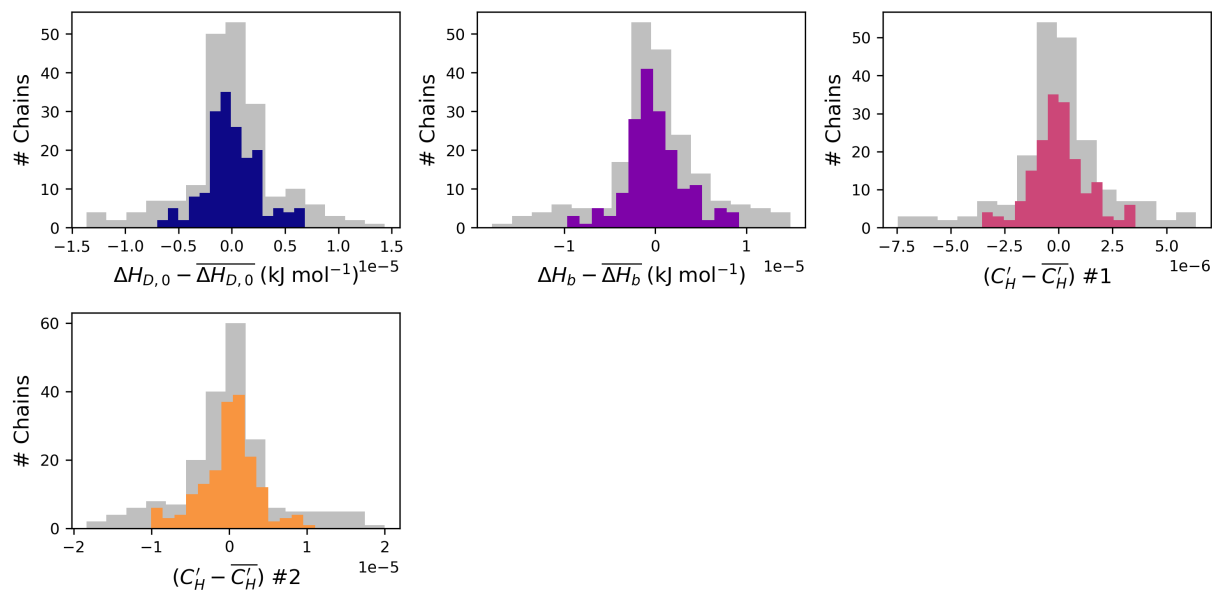


Linear Free Energy Relationships

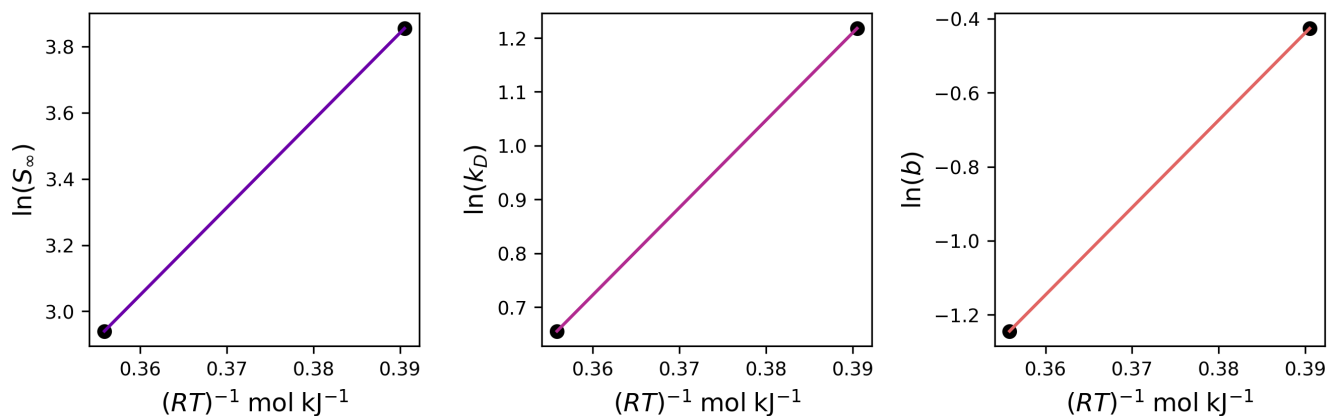
Heat of Henry Sorption	$\Delta H_D = -2.8437 + 2.6143 \ln(k_{D,0})$
Heat of Langmuir Sorption	$\Delta H_b = 1.6629 + 2.62 \ln(b_0)$



van't Hoff Relationships



Energetics



	Entropic Prefactor ($S_{i,0}$)	ΔH_i (kJ/mol)
i = Infinite Dilution	$1.584\text{e-}03 \pm \text{inf}$	$-26.382164 \pm \text{inf}$
i = Langmuir	$6.035\text{e-}03 \pm \text{inf}$	-16.203309 ± 0.941819
i = Henry	$6.481\text{e-}05 \pm \text{inf}$	-23.604591 ± 0.276522

Analysis Settings

Parameter	Value
ΔH_D Initial Guess Range	[-1, -30]
ΔH_b Initial Guess Range	[-1, -30]
$k_{D,0}$ Initial Guess Range	[0.001, 0.01]
b_0 Initial Guess Range	[0.0001, 0.005]
ΔH_D Solver Bounds	[-50, 0]
ΔH_b Solver Bounds	[-50, 0]
$k_{D,0}$ Solver Bounds	[0, None]
b_0 Solver Bounds	[0, None]
C'_H #1	[0 150]
C'_H #2	[0 150]
Trials	200
LFER Solver	SLSQP
maxiter (LFER solver)	1000
van't Hoff Solver	SLSQP
maxiter (van't Hoff solver)	1000
ftol (SLSQP)	1e-12
xtol (trust-constr)	1e-12
gtol (trust-constr)	1e-12

Questions/Comments/Concerns?

Documentation available through ReadTheDocs (*)

Please reach out to Brandon C. Tapia via email (bctapia@mit.edu) or through the pyDMS GitHub page

Code Availability

pyDMS can be downloaded via:

- GitHub (source code): git clone ****(insert link)**
- pip: pip install ****insert name**
- Anaconda: conda install ****insert name**

Citation

If you used pyDMS in any presented or published work please cite:

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